PARDHU SRI RUSHI VARMA KONDURU

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SUMMARY

Final-year B.Tech student focused on systems security and machine learning, with two first-author preprints (Kernel Borderlands, RLAE) and hands-on experience in eBPF, Linux internals, and cloud deployment. Builds and operates systems end to end, from kernel-level detection to production infrastructure.

EDUCATION

Malla Reddy University, Hyderabad **Aug 2023 – Apr 2027**
B.Tech, Computer Science & Engineering (Cyber Security) *CGPA: 7.73 / 10*
Coursework: Operating Systems, Computer Networks, Data Structures, Computer Architecture.

TECHNICAL SKILLS

- **Languages:** Python, Go, Rust, C/C++, TypeScript/JavaScript, Bash
- **Systems & Kernel:** Linux internals, eBPF (libbpf, BPF CO-RE), libalpm, pacman/makepkg, bubblewrap sandboxing, embedded firmware (ESP32)
- **Cloud & DevOps:** GCP, AWS, Docker, Kubernetes, CI/CD (GitHub Actions), Nginx, Linux, networking, Git, zero-downtime rollouts, IaC
- **Security:** VAPT, threat modeling, multi-cloud red teaming, CTF challenge design, container security, secrets & access control, binary hardening, supply-chain auditing, incident response
- **Backend & Web:** Node.js, REST APIs, WebSocket, PostgreSQL, MongoDB, React, Vue, Puppeteer, MCP, Git
- **Data & AI:** PyTorch, scikit-learn, HMM/PCA/UMAP drift detection, multi-agent RL, LLM/GenAI concepts, Jupyter

SYSTEMS & EMBEDDED PROJECTS

- **Kernel Borderlands — Kernel-Level Runtime Defense Framework** *(C/eBPF, Go, Python, Rust — Project Lead)* — Led a four-person team and authored most of the codebase: detects, classifies, and contains anomalous process behavior at Ring 0 using eBPF telemetry, behavioral analytics, and multi-agent response. Six eBPF programs (C, libbpf, CO-RE) feeding a ring buffer into a userspace scoring service; Go control plane (`kbd`); Python multi-agent RL defense swarm; Rust script-safety checker (`kb-checker`); React dashboard, Bubble Tea TUI, and MCP server.
- **Emrac — Safety-First Hybrid Package Manager for Arch Linux** *(Rust, libalpm, pacman/makepkg, bubblewrap)* — Orchestration layer over pacman, makepkg, and libalpm unifying official repos and the AUR in one CLI; core loop complete, 30+ commands, container-tested, no telemetry; 53-feature spec. Safe upgrades (plan digests, pre-flight checks, update risk rating, Arch news gate); security and recovery (static PKGBUILD/package audit, AUR trust signals, bubblewrap-sandboxed builds, hash-chained history, tamper-evident ledger, undo/rollback/downgrade/snapshot); tooling (dependency tree/impact/cycles, doctor, pacnew, cache, rebuild, pin, self-upgrade).
- **PhishVault** *(Node.js, MongoDB, Puppeteer, Docker)* — Containerized URL analysis, classification, and reporting platform.
- **OuroMini** *(C++, Vue)* — ∼300 KB firmware ported to display-less ESP32 in 15 days; WebSocket API + dashboard.

EXPERIENCE

Breach Point CTF 2026 — National Capture-the-Flag Event **Dec 2025 – Oct 2026**
Core Team Member, Lead Technical Challenge Author & Deployment Operations *Malla Reddy University / GCP*

- Authored **21 challenges** across two seasons (Siege of Troy, Federation) in 12 categories (web, API, reverse engineering, pwn, IoT/OT, blockchain, crypto, stego, OSINT), Easy to Hard++; each ships as a Docker Compose stack with a per-instance flag injected in memory at boot (zenrage.pardhuvarma.tech).
- Owned end-to-end cloud deployment and DevOps for a 36-hour hybrid event serving **321 concurrent teams** (incl. international), latest Oct 3–4, 2026: parallel GCP deployments, Docker, CI/CD, zero-downtime rollout, live troubleshooting.
- Built Git-driven build → test → deploy → monitor workflows (GitHub Actions status board); hardened infrastructure: scoped container capabilities instead of privileged mode, per-challenge resource limits, isolation, log review, rollbacks.

Independent Research **Dec 2025 – Present**
Systems Security & ML Research — Independent Researcher *Remote*

- **Kernel Borderlands** — first-author preprint (Preprints.org 202610.2818, Oct 2026), *An eBPF-Based Behavioral Detection and Multi-Agent Containment Architecture for Linux Server Workloads*; four-author paper on the kb-core design, enforcement and multi-agent response.
- First-author preprint (arXiv:2603.02934) on reversible, auditable AI adaptation; LBSM manuscript in prep.: telemetry pipelines modeling system behavior as statistical manifolds (HMM, PCA/UMAP, drift detection).

HONORS & AWARDS

2nd Place, GrowthNinja Hackathon — IIT Hyderabad (2025) • **15th/50**, CY4TF Cyber Fortress CTF — Team Captain (2025) • Certs: MCRTA, ADRTS, CRTID (CWLabs), Palo Alto Networks Cybersecurity Fundamentals